

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Data Engineering professional with **5+ years of experience** designing, building, and optimizing production-grade data solutions using Python, T-SQL, Microsoft Azure, Azure Data Factory, Synapse, Microsoft Fabric, Databricks, and modern Lakehouse architectures.
- Hands-on experience developing scalable **ETL/ELT pipelines, Fabric Data Pipelines, notebooks, lakehouses, and warehouse solutions**, transforming raw enterprise data into trusted, analytics-ready assets for reporting, advanced analytics, and AI use cases.
- Strong expertise in **Microsoft Fabric, OneLake, Power BI DirectLake, Delta Lake, medallion architecture, dimensional modeling, semantic modeling, relational design, data transformation, and performance tuning**.
- Experienced integrating data from **SaaS, ERP, APIs, relational databases, cloud storage, Dynamics 365, Workday, and enterprise applications**, applying secure and reusable ingestion and orchestration patterns.
- Implemented **automated data-quality checks, lineage, observability metrics, reconciliation, security controls, and governance standards** to improve reliability, traceability, and trust across enterprise data products.
- Strong DevOps experience with **Azure DevOps, GitHub, YAML pipelines, CI/CD, Terraform, Bicep, Git, deployment automation, monitoring, and infrastructure-as-code practices** across cloud data platforms.
- Experienced in client-facing consulting environments, collaborating with **solution architects, analysts, business stakeholders, and engineering teams** to gather requirements, demo iterative deliverables, document technical designs, explain complex concepts, and deliver scalable data solutions.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Querying:** Python, T-SQL, SQL, PySpark, Spark SQL, Java, Pandas, Notebook Development, Data Wrangling, Data Transformation, Performance Tuning.
- **Microsoft Data Stack:** Microsoft Fabric, OneLake, Fabric Lakehouse, Fabric Data Pipelines, Fabric Notebooks, Fabric Warehouse, Power BI DirectLake, Azure Data Factory, Azure Synapse Analytics, Synapse Pipelines.
- **Lakehouse & Data Engineering:** Databricks, Delta Lake, ETL/ELT, Medallion Architecture, Bronze/Silver/Gold Layers, Data Ingestion, Data Orchestration, Data Modeling, Relational Modeling, Dimensional Modeling, Semantic Models.
- **Data Quality & Governance:** Automated Data Quality Checks, Data Validation, Reconciliation, Data Lineage, Observability Metrics, Data Governance, Security, Metadata, Monitoring, Reliability, Cost Optimization.
- **Sources & Integration:** SaaS/ERP Integration, Dynamics 365, Workday, Costpoint Concepts, REST APIs, Relational Databases, SQL Server, PostgreSQL, MySQL, Oracle, Cloud Storage, JSON, CSV, Parquet.
- **DevOps & Infrastructure:** Azure DevOps, GitHub, Git, YAML Pipelines, CI/CD, Terraform, Bicep, Docker, Infrastructure as Code, Deployment Automation, Release Management, Environment Configuration.
- **Consulting & Delivery:** Requirements Gathering, Client Workshops, Solution Design, Technical Documentation, Iterative Demos, Stakeholder Communication, Reusable Accelerators, Performance Optimization, Agile/Scrum, Continuous Improvement.

EXPERIENCE

Software Engineer AI PCMI LLC (Full Time)

Oct 2024 – Present
Dallas, TX, USA

- Designed and delivered enterprise **Lakehouse and analytics solutions** using Microsoft Fabric, Azure Data Factory, Databricks, Python, T-SQL, PySpark, and Delta Lake, transforming insurance data into trusted datasets for reporting, analytics, and AI-driven use cases.
- Built scalable **ETL/ELT and Fabric Data Pipelines** that ingested data from enterprise applications, REST APIs, SQL databases, cloud storage, and SaaS platforms into OneLake and medallion-style Bronze, Silver, and Gold layers.
- Developed reusable **Python, PySpark, T-SQL, and notebook-based transformations** for data cleansing, enrichment, aggregation, reconciliation, and dimensional modeling while optimizing joins, queries, partitions, and processing costs.
- Designed **relational, dimensional, and semantic data models** supporting business entities, historical transactions, analytics-ready datasets, Fabric Warehouse, Power BI DirectLake, and downstream reporting consumption.
- Implemented **automated data-quality controls, lineage, observability metrics, governance, security, and reconciliation frameworks** to validate completeness, accuracy, consistency, and traceability across production pipelines.
- Implemented DevOps practices using **Azure DevOps, GitHub, YAML CI/CD pipelines, Terraform, Bicep, Git, monitoring, and automated deployment workflows**, improving release consistency and operational support across data environments.
- Collaborated with **solution architects, analytics engineers, product owners, and business stakeholders** to gather requirements, present iterative demos, document technical designs, resolve performance issues, and create reusable accelerators for broader platform adoption.

Environments: Microsoft Fabric, OneLake, Fabric Data Pipelines, Fabric Notebooks, Fabric Warehouse, Power BI DirectLake, Azure Data Factory, Azure Synapse, Databricks, Delta Lake, Python, T-SQL, PySpark, ETL/ELT, Medallion Architecture, Data Quality, Lineage, Azure DevOps, GitHub, YAML, Terraform, Bicep, CI/CD.

Software Engineer AI Client: CVS

Dec 2023 – Sept 2024
Remote, USA

- Designed and developed healthcare **data engineering and Lakehouse solutions** using Python, T-SQL, PySpark, Azure Data Factory, Databricks, and Delta Lake to support analytics, operational reporting, and enterprise data consumption.
- Built production **ETL/ELT pipelines** integrating healthcare data from REST APIs, relational databases, SaaS applications, files, and cloud storage into standardized raw, refined, and curated data layers.
- Developed notebook and Spark-based transformations using **Python, PySpark, SQL, Delta tables, data wrangling, cleansing, aggregation, and enrichment**, improving data usability and processing efficiency across large datasets.
- Designed **dimensional and semantic data models** for analytics and business intelligence, supporting fact/dimension structures, historical reporting, curated data products, and Power BI-oriented consumption patterns.
- Implemented **data-quality checks, reconciliation, lineage, observability, security, and governance controls** to identify missing records, schema issues, duplicate data, inconsistencies, and upstream/downstream processing failures.
- Integrated data solutions with **Azure DevOps, Git, YAML pipelines, CI/CD, Terraform concepts, monitoring, logging, and deployment automation**, supporting reliable promotion across development and production environments.
- Partnered with architects, analysts, data engineers, and business stakeholders to **gather requirements, demonstrate iterative deliverables, explain technical designs, optimize pipelines, and translate business needs into scalable data solutions**.

Environments: Python, T-SQL, PySpark, Azure Data Factory, Databricks, Delta Lake, ETL/ELT, Data Pipelines, REST APIs, SQL Server, PostgreSQL, SaaS Integration, Dimensional Modeling, Data Quality, Lineage, Azure DevOps, YAML Pipelines, Git, CI/CD, Power BI Concepts, Monitoring.

Software Engineer Infor

Sept 2021 – Dec 2022
Hyderabad, India

- Designed and developed enterprise **data pipelines and analytical solutions** using Python, T-SQL, PySpark, SQL, and cloud data technologies, supporting high-volume operational applications and reporting workloads.
- Built reusable **ETL/ELT ingestion and orchestration workflows** to extract data from relational databases, APIs, ERP applications, and files and transform raw information into curated warehouse and analytics-ready datasets.
- Developed scalable data transformations using **Python, PySpark, SQL, notebook-based processing, data cleansing, enrichment, aggregation, and reconciliation**, improving data consistency and processing performance.
- Designed **relational and dimensional data models** using fact and dimension structures, surrogate keys, historical attributes, semantic concepts, and optimized SQL patterns for downstream analytics and BI consumption.
- Implemented **data-quality validation, lineage concepts, governance controls, monitoring, and troubleshooting** to improve reliability, traceability, and operational support across integrated enterprise data workflows.
- Applied software delivery practices using **Git, Jenkins, Azure DevOps concepts, CI/CD, YAML concepts, Docker, deployment automation, and technical documentation** to standardize development and release processes.
- Collaborated with solution architects, analysts, product teams, and stakeholders to **analyze requirements, document technical designs, optimize SQL workloads, demonstrate solutions, and contribute reusable components and engineering best practices**.

Environments: Python, T-SQL, PySpark, Apache Spark, SQL Server, PostgreSQL, MySQL, Oracle, ETL/ELT, ERP Integration, REST APIs, Data Warehousing, Dimensional Modeling, Data Quality, Git, Jenkins, Azure DevOps Concepts, CI/CD, Docker, Monitoring, Technical Documentation.

Associate Engineer

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed enterprise **data integration and transformation solutions** using Python, T-SQL, SQL, and relational databases, supporting business reporting, operational analytics, and downstream application requirements.
- Built reusable **ETL ingestion pipelines** to extract information from ERP systems, REST APIs, databases, flat files, and enterprise applications and transform source data into standardized analytical datasets.
- Implemented Python and SQL transformations for **data wrangling, cleansing, validation, aggregation, enrichment, and reconciliation**, improving accuracy and consistency across recurring data-processing workflows.
- Designed and maintained **relational and dimensional models** using PostgreSQL, MySQL, fact/dimension structures, surrogate keys, normalized tables, and reporting-oriented schemas for reliable data consumption.
- Applied **data-quality, governance, security, monitoring, and documentation practices** to identify missing records, duplicate information, schema inconsistencies, integration failures, and other data reliability issues.
- Supported engineering delivery through **Git, CI/CD, deployment workflows, Linux, logging, troubleshooting, version control, and environment management**, improving repeatability and maintainability of enterprise data solutions.
- Worked with architects, analysts, engineers, and business stakeholders to **gather requirements, translate business concepts into technical solutions, communicate findings, resolve data issues, and contribute reusable integration patterns**.

Environments: Python, T-SQL, SQL, PostgreSQL, MySQL, ETL, Data Ingestion, Data Transformation, ERP Integration, REST APIs, Data Modeling, Dimensional Modeling, Data Quality, Data Governance, Git, CI/CD, Linux, Logging, Monitoring, Technical Documentation.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.